

Artificial Intelligence for Predicting Treatment Adherence in Opioid Use Disorder: A Scoping Review

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Background and Aim

Opioid use disorder (OUD) is a chronic condition in which an individual engages in the persistent use of opioids that causes significant distress and negatively impacts their societal functioning. Treatment for OUD involves pharmacological therapies such as methadone, buprenorphine, and naltrexone, typically used in combination with behavioral interventions such as counselling and cognitive behavioural therapy. However, non-adherence to OUD treatment is high, potentially leading to negative outcomes like relapse and increased risk of overdose. Therefore, identifying patients at risk of treatment nonadherence is essential to ensure that OUD is adequately managed. Models utilizing AI and ML techniques have emerged as promising candidates to achieve risk stratification in this patient population. We conducted a scoping review to capture and systematically map existing literature on AI and ML applications predicting adherence to treatment in individuals with OUD.

Methods

Ovid MEDLINE, Embase, PsycINFO, Web of Science, Scopus, CINAHL, IEEE Xplore, and ACM Digital Library were searched to identify and obtain peer-reviewed empirical research articles published from inception to October 7, 2024. Twenty-two studies were selected to be included in the review.

Results

All studies that matched our search criteria were published after 2018 and predominantly conducted in the United States. Random forest models were frequently identified as the top performer although significant variability in algorithms, evaluation metrics, and key predictors was noted in the literature.

Conclusion

The need for future research to cover more geographical locations, diversify patient populations, focus on a standardized group of models and outcomes, and utilize larger samples was highlighted.

Keywords: Machine learning; Outcome prediction; Persistent opioid use; Substance abuse; Substance misuse

Introduction

Opioid use disorder (OUD) is a chronic condition characterized by a prolonged, problematic pattern of opioid usage leading to significant distress or impairment in everyday life [1]. OUD affects more than 16 million people globally and claims over 120,000 lives per year [1]. According to the 2023 U.S. National Survey on Drug Use and Health, 8.6 million people aged 12 years or older in the United States had misused prescription opioids in the preceding year [2]. It is difficult to accurately determine OUD prevalence as national surveys often exclude high-risk populations such as homeless, hospitalized, or incarcerated individuals [3]. Nevertheless, recent estimates have indicated stability or even a small decline in the prevalence of OUD in the United States [3]. Despite a decrease in prescription opioid prescribing in the US over the past decade, over 6 million people in the country continue to have OUD, heroin use has substantially increased, and synthetic opioid-related mortality constitutes a major cause of death [3, 4]. Furthermore, individuals with untreated OUD often experience challenges in their physical, emotional, and social functioning, leading to a decline in their quality of life [5, 6]. Therefore, it is important that OUD is managed effectively.

OUD management is typically pharmacological and involves the routine administration of methadone, buprenorphine, or naltrexone as the gold standard [7-10]. Pharmacological treatment for OUD reduces symptoms of opioid withdrawal, decreases cravings, attenuates risk of relapse, and lowers the likelihood of overdose mortality while improving psychosocial functioning [7, 10]. OUD may also be managed using behavioural interventions, such as counselling, motivational interviewing, cognitive behavioural therapy, acceptance and commitment therapy, relational psychotherapy, and contingency management [8, 11]. These interventions are often used in conjunction with medication and are aimed at addressing the psychosocial elements of OUD, increasing treatment compliance, minimizing drug relapse, and managing comorbid psychiatric disorders [8, 11].

The effectiveness of treatments for OUD is limited due to relatively high rates of nonadherence [12]. Nonadherence among individuals with OUD is associated with poor outcomes, including an increased risk of relapse and overdose as well as greater healthcare costs [12-14]. Reducing rates of nonadherence to treatment for OUD is thus of prime importance. This requires an accurate assessment of the likelihood of treatment adherence and a subsequent identification of individuals that are at risk, allowing for timely mitigation of any challenges or barriers that might be keeping them from continuing their treatment. Models that leverage machine learning (ML) and artificial intelligence (AI) can be used to assist in predicting adherence to OUD treatment. ML algorithms are a branch of AI that uses big data as input to identify patterns and compute predictions for outcomes of interest [15-18]. The benefit of ML approaches lies in the ability of models to learn or train on input data to continually improve their predictive performance [16-18].

Although several studies have explored the use of ML and AI in predicting adherence to treatment for OUD, attempts to synthesize and capture the landscape of research on the topic have not been made. This scoping review was conducted to provide an overview of different types of ML and AI applications used to predict OUD treatment adherence and evaluate their effectiveness and accuracy to identify models with the best predictive performance.

Methods

I. *Approach and framework*

Scoping reviews are a type of evidence synthesis that are considered the methodology of choice for comprehensively capturing the landscape of research on a topic, highlighting any gaps in knowledge [19]. This methodology was consistent with the objective of synthesizing extant literature on predicting OUD treatment adherence using ML and AI algorithms.

This scoping review was conducted in accordance with the Arksey and O'Malley (2005) framework which consists of 6 stages, including 1) developing a research question; 2) conducting a literature search; 3) screening studies against an inclusion and exclusion criteria; 4) extracting data from included articles; 5) analyzing abstracted data; and 6) a stakeholder consultation. The final step of the framework is considered optional [20] and thus, was not performed as a part of this review. The structure and content of this scoping review was guided by the preferred reporting items for systematic reviews and meta-analyses extension for scoping reviews (PRISMA-ScR) [21]. The protocol for this review was registered on Open Science Framework (OSF) [22].

II. *Search strategy, study selection, and screening*

A comprehensive search strategy was developed using keywords such as artificial intelligence, opioid, treatment adherence, and compliance that would capture articles on the use of ML and AI models in predicting treatment adherence for OUD. The search was run across Ovid MEDLINE, Embase, PsycINFO, Web of Science, Scopus, CINAHL, IEEE Xplore, and ACM Digital Library from inception to October 7, 2024. The full search strategy is present in the supplementary material.

Articles obtained from the search results were uploaded onto Covidence (Veritas Health Innovation, Melbourne, Australia; available at www.covidence.org). The title and abstracts of the retrieved studies were reviewed to determine their eligibility against the inclusion criteria. Studies that were included after title and abstract screening were further assessed through a full-text review. Articles were screened independently by two reviewers at both screening stages and disagreements were resolved by a third reviewer. Peer-reviewed, primary and secondary empirical studies on ML and AI applications to predict treatment adherence in individuals with

LOUD were deemed eligible. Relevant conference papers were included only if they were from IEEE or ACM. There were no restrictions on study setting, date of publication, or language. Articles were excluded if they did not include predictive ML and AI models or assess outcomes of interest. Moreover, studies with non-human participants were considered ineligible. Non-research articles (e.g., erratums, commentaries, book chapters, letters to the editor, and protocols), thesis dissertations, and reviews were also eliminated.

III. Data extraction and analysis

Data extraction was completed on Covidence using a standardized template populated independently by a pair of researchers for each study. The abstracted information was reviewed for accuracy by a third researcher and conflicts were resolved through consensus. The following information was collected: 1) title of the article; 2) primary author; 3) publication date; 4) aims of the study; 5) country where the study was conducted; 6) study design; 7) location or site; 8) sample size; 9) sociodemographic characteristics (e.g., age, gender, race or ethnicity, medical history or comorbidities); 10) description of outcomes; 11) how outcomes were measured (e.g., surveys, health data records, questionnaires); 12) length of data collection; 13) description of treatment intervention; 14) ML and AI models discussed; 15) source of input data (e.g., ECG, electronic health records); 16) predictive or modelling variables; 17) size of training and testing dataset; 18) performance metrics (e.g., 5-fold cross validation, random sampling); 19) best model and its evaluation metrics (e.g., accuracy, precision, recall, sensitivity, specificity, positive predictive value); 20) key findings of the study; and 21) its limitations. The extracted data was analyzed and reported using a narrative synthesis approach.

Results

Search results

The database search yielded 2800 hits. Of these 2729 articles were removed at the title and abstract screening stage with another 49 articles excluded following full-text review. A total of 22 articles were deemed eligible and included in this scoping review (Figure 1).

Results of individual sources of evidence

The study characteristics and main findings of each of the 22 included articles are summarized in Tables 1-2.

Synthesis of findings by year

We identified an increase in the number of studies assessing the application of ML and AI models to predict treatment adherence among individuals with OUD over time (Table 3). The smallest number of studies that matched our search criteria were published from 2018 to 2020, each of which had only one (4.5%) relevant publication [29, 41, 43]. Three (13.7%) studies were published in 2021 [27, 33, 38], another three (13.7%) in 2022 [25, 31, 42], and four (18.2%) in 2023 [26, 28, 39, 44]. A majority of the included articles, amounting to nine (40.9%) studies, were published in 2024 [23, 24, 30, 32, 34-37, 40].

Synthesis of findings by country

An analysis of included articles by country showed that most studies were conducted in the United States (Table 4). Numerically, 20 (91%) out of the 22 studies in this scoping review were US-based [23-26, 28-35, 37-44]. Only two of the included studies were based outside of the United States. Of these remaining studies, one (4.5%) was conducted in Australia [27] and one (4.5%) in Taiwan [36].

Synthesis of findings by adherence outcomes

Induction in OUD treatment [23], opioid cessation [29], relapse [37], and patient satisfaction with OUD treatment [39] were the least frequently explored outcomes and were assessed in only one (4.5%) study each. These were followed by non-adherence with treatment, which was the examined outcome in two (9.1%) articles [38, 41]. Three (13.7%) studies used AI and ML algorithms to predict treatment engagement and adherence [28, 30, 44], while retention in OUD treatment programs constituted the outcome of predictive models in four (18.2%) articles [26, 32, 34, 36]. The second most frequently examined outcome was treatment completion. This was assessed in six (27.3%) of the 22 studies in this scoping review [25, 26, 35, 38, 40, 43]. Treatment discontinuation, cessation, or drop out was explored in seven (31.8%) articles, making it the most frequently measured outcome in literature on the application of AI and ML algorithms to predict adherence to treatment and management among patients with OUD [24, 27, 31, 33, 36, 38, 42]. The number of included studies by outcome are presented in Table 5.

Synthesis of findings by best reported AI and ML models

Random forest models were identified as the best performing AI and ML algorithms for predicting treatment adherence outcomes among individuals with OUD in ten (45.5%) studies, constituting a majority of the included articles [24-26, 30-33, 40-42]. Chi-square automatic interaction detection decision tree [35, 44], extreme gradient boosting [28, 38], and support vector machine [29, 43] were deemed the most effective predictive models in two (9.1%) studies each. Of the remaining six articles, one (4.5%) study each identified the following as top performing AL and ML models: deep learning [27], gradient boosting machine [34], logistic regression [37], neural network [36], ridge regression [39], and Tian-EN [23]. A list of AI and ML models and the corresponding number of studies that report them as having the highest predictive power is displayed in Table 6.

Discussion

This scoping review was conducted to provide an overview of the state of evidence on the use of AI and ML models to predict outcomes related to treatment adherence among patients with OUD. The findings from each of the 22 included articles were synthesized individually and reported along with information on the studied AI and ML algorithms, their predictive performance, key modelling variables, and the adherence outcomes measured. Moreover, studies were collectively examined to identify patterns in the context of publication dates, geographic distribution, treatment adherence outcomes, and top AI and ML models.

The included studies were all published between 2018 and 2024 [23-44], indicating that predicting adherence to treatment among patients with OUD using AI and ML algorithms is a fairly recent topic within the research literature. Relevant articles were found to be low in number during the first three years [29, 41, 44]. The trend changed in 2021 with an increasing number of studies being published each year [23-28, 30-40, 42, 44], potentially representing a rising interest in the topic.

Geographically, evidence on the application of AI and ML tools to forecast treatment adherence in an OUD patient population comes from three countries: Australia, Taiwan, and the United States, all of which are considered developed economies [23-44]. Only one (4.5%) of the included studies was based in Taiwan (i.e., the east) [36], while an overwhelming majority (21 studies, 95.4%) of the relevant research was concentrated in western countries [23-35, 37-44]. This illustrates a lack of representation of both developing and eastern countries in pertinent literature, in turn limiting the generalizability of findings to these regions. Moreover, a major proportion (20 articles, 91%) of the studies in this scoping review were conducted in the United States [23-26, 28-35, 37-44]. It is possible that the findings in these articles regarding top models and predictors do not extend to a non-American context, impacting the generalizability of these studies and ultimately the results of this scoping review.

Although random forest models took an obvious lead in the proportion of articles identifying them as the best algorithm [24-26, 30-33, 40-42], there was substantial heterogeneity across studies regarding which AI or ML technique was most suitable for the analyses. However, candidate AI models and applications assessed in the literature were similar. Random forest algorithms were an example of one such model and were evaluated in 14 (63.6%) articles [24-26, 29-34, 38-42]. Another example included models employing logistic regression which were examined in 11 (50%) studies [24-26, 28, 31-34, 37, 39, 41]. Similarly, a comparative analysis of the predictive ability of extreme gradient boosting was performed as a part of eight (36.4%) studies [24, 25, 28, 32, 33, 38-40]. While many algorithms were tested in several of the included publications, some candidate models appeared much less frequently. For example, the performance of artificial neural networks [33, 38] and chi-square automatic interaction detection decision trees [35, 44] were investigated in only two (9.1%) studies. Similarly, models like Gentle Boost and LogitBoost were explored in a single (4.5%) study. These observations point towards a need for additional evidence, preferably with a more standardized set of candidate models, to provide deeper insight into which models excel in their ability to forecast treatment adherence in individuals with OUD relative to other algorithms.

Heterogeneity in the data was also evident in the evaluation metrics used to gauge the performance of AI and ML models. Performance was assessed across a major (nine studies, 40.9%) proportion of the included articles with AUC [24-26, 30, 35, 38-41]. Accuracy was utilized as a metric in seven (31.8%) studies [25, 26, 28, 29, 30, 41, 43]. Precision [25, 32, 33, 38, 40], recall [25, 32, 33, 38, 40], and specificity [28, 30, 31, 40, 43] were employed as performance metrics in five (22.7%) publications. This was followed by AUROC [28, 34, 37, 42], F-1 score [25, 28, 33, 39], and sensitivity [28, 30, 31, 43] were calculated in four (18.2%) of the 22 articles. Other measures of evaluation included c-statistics [24, 27, 33], positive and negative predictive values [28, 43], qini-value [23], c-for-benefit [23], d-statistics [27], r-squared value [27], integrated Brier score [27], AUPR [28], mean correlation coefficient [36], and balanced accuracy [38]. The heterogeneity seen in performance metrics limits cross-study comparisons, particularly for measures that are used in few studies, making it difficult to draw meaningful conclusions.

Several of the five most important modelling features were common to at least two studies. This included age, employment levels, geographic location, length of stay, service setting, source of referral, and involvement with the criminal justice system [24-27, 29, 30, 35, 38, 40, 42]. This overlap among modelling variables suggests that AI and ML models from these studies might potentially be applicable to different samples or populations of patients while retaining their predictive performance. However, substantial research is required into the external validity of the AI and ML models before their external validity can be conclusively established. It is also worth noting that top predictive variables were not reported in seven (31.8%) of the 22 articles [32-34,

36, 39, 41, 43], hindering attempts at testing for replicability by researchers outside of the original group of authors.

Moreover, the findings of this scoping review must be interpreted in light of some additional limitations. First, there was a significant degree of variation in the outcomes used to assess treatment adherence. Second, the adherence-related outcomes in the included studies are measured over a duration of one year or less. As such, the best reported algorithms identified in this review may not be as effective in predicting longer-term outcomes, potentially affecting the generalizability of the results. Third, some studies tested their AI and ML models on a small sample. The results from these studies had low statistical power. Fourth, despite the comprehensiveness of the search strategy, there remains a possibility of the exclusion of potentially eligible studies that may not have been indexed in the selected databases. Finally, there is a possibility that factors relevant for AI and ML algorithms might evolve over time, resulting in changes to the optimal set of predictive variables and models considered most suitable for predicting adherence outcomes in an OUD patient population. This implies that the scoping review may need to be updated in a few years to ensure that the findings are reflective of changes in the relevant literature.

Conclusion

OUD is a significant public health and societal concern, often associated with poor treatment adherence. This scoping review synthesized literature on leveraging AI and ML to predict treatment adherence and the lack thereof in patients with OUD. An interest surrounding the topic was on the rise since 2018 with most of the research coming from the United States and random forest algorithms taking the lead as the model of choice in terms of predictive performance. Moreover, there was considerable heterogeneity in the top models, performance metrics, and modelling variables, highlighting the need for more standardized models, outcomes, and predictive variables in future research to facilitate cross-study comparisons.

Author Contributions

Aghna Wasim, Ali Abud, and Samir Malick: investigation, methodology, conceptualization, screening, data extraction, writing (original draft), and writing (review & editing); Nazeefa Arifina Nashrah, Veronika Lošanová, Hetvi Raimugia, Marisha Karim, Minh Van Khanh Le, Venussha Baskarathasan, Charlotte Gibson, Siba Alkhatib: screening, data extraction, writing (review & editing); Silvia S. Martins: critical revision of the manuscript, writing (review & editing). All authors have read and approved the manuscript for submission.

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Declaration of Interests

The authors declare that they have no competing interests.

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Data Availability Statement

All data utilized in this study are available in the published article or the supplementary material.

Supporting Information

Supplementary Material 1: Complete List of Predictive or Modeling Variables for Each Study and Search Strategy

Supplementary Material 2: PRISMA-ScR Checklist

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Table Legends

Table 1. Summary of study characteristics

Table 2. Summary of AI/ML models utilized

Table 3. Number of included studies per year

Table 4. Number of included studies per country

Table 5. Number of included studies per treatment adherence outcome

Table 6. Number of included studies per best reported model

Figure Legends

Figure 1: Preferred Reporting Items for Scoping reviews and Meta-Analysis (PRISMA) flow chart for the study

Table 1. Summary of study characteristics

Author (Year of Publication)	Title	Study Aim(s)	Country	Study Design	Sample Size; Female (%)	Outcome of Interest
Afshar et al. (2024) [22]	Machine learning–driven analysis of individualized treatment effects comparing buprenorphine and naltrexone in opioid use disorder relapse prevention	Optimize treatment decisions and customize them for individual patients profile to improve effectiveness of treatment and clinical outcomes	USA	Secondary analysis of a multicenter randomized trial	570; 169 (29.6%)	Induction of outpatient treatment with opioid-relapse-free survival for 24 weeks
Al Faysal et al. (2024) [23]	An explainable machine learning framework for predicting the risk of buprenorphine treatment discontinuation for opioid use disorder among commercially insured individuals	Develop and evaluate a machine learning framework for predicting buprenorphine care discontinuity within 12 months following treatment initiation	USA	Retrospective cohort	30373 ; 13106 (43.15%)	Treatment discontinuation (gap of ≥ 30 days without a buprenorphine prescription) within 12 months of treatment initiation

Baird et al. (2022) [24]	Use of machine learning to examine disparities in completion of substance use disorder treatment	Examine disparities in the completion of substance use disorder treatment in the USA	USA	Counterfactual research design	649479; 246744 (38%)	Treatment completion
Baird et al. (2023) [25]	Determinants of outpatient substance use disorder treatment length-of-stay and completion: the case of a treatment program in the southeast U.S	Apply machine learning specifically, virtual twins to determine which factors, in what combination, and at what levels were most predictive of longer lengths-of-stay in treatment and successful treatment completion	USA	Observational prognostic	256; NR	Length-of-stay (>90 days) Successful treatment completion (did not leave against medical advice or for other reasons, such as extended incarceration)
Bharat et al. (2021) [26]	Using administrative data to predict cessation risk and identify novel predictors among new entrants to opioid agonist treatment	Develop and compare traditional and deep learning-extensions of Cox regression to examine the potential for predicting time in	Australia	Retrospective cohort	16576 ; 5072 (30.6 %)	12-month opioid agonist treatment cessation

		opioid agonist treatment at individuals' first episode entry				
Burgess-Hull et al. (2023) [27]	Using machine learning to predict treatment adherence in patients on medication for opioid use disorder	Develop and validate an automated risk-modeling framework to predict opioid abstinence and medication adherence at a patient's next attended appointment and evaluate the predictive performance of machine learning algorithms versus logistic regression	USA	Retrospective cohort	2742; 1152 (42%)	Treatment adherence at patient's next appointment
Cox et al. (2020) [28]	Identifying factors associated with opioid cessation in a biracial sample using machine learning	Apply multiple machine learning techniques to evaluate a large set of clinical, demographic, general health, and behavioral variables in a large, racially mixed cohort of	USA	Cohort	3749; 1365 (36.4%)	Opioid cessation (>1 year since opioid use)

		individuals to identify factors that are associated with opioid cessation				
Eddie et al. (2024) [29]	Exploring predictors of substance use disorder treatment engagement with machine learning: The impact of social determinants of health in the therapeutic landscape	Explore associations between individual- and neighborhood-level factors, and SUD treatment engagement	USA	Secondary analysis of the Global Appraisal of Individual Needs (GAIN) dataset and United States Census Bureau data	15873 ; 4278 (27.0 %)	Treatment engagement at 3-month follow up
Gottlieb et al. (2022) [30]	Machine learning for predicting risk of early dropout in a recovery program for opioid use disorder	Introduce a machine learning model to a retrospective database of over 700 enrolled patients to identify individuals at higher risk of	USA	Retrospective cohort	90-day analysis = 715; NR 120-day	Opioid use disorder recovery program dropout at 90 and 120 days

		dropping from the program after 90 and after 120 days			analysis = 691; NR	
Haredasht et al. (2024) [31]	Predictability of buprenorphine-naloxone treatment retention: A multi-site analysis combining electronic health records and machine learning	Determine the predictability of 6-month retention in buprenorphine-naloxone treatment using electronic health record data from diverse clinical settings and to identify key predictors	USA	Retrospective observational	9757; 4371 (44.8%)	6-month retention in buprenorphine-naloxone treatment (continuous prescription for at least 6 months without a gap of >30 days)
Hasan et al. (2021) [32]	A machine learning based two-stage clinical decision support system for predicting patients' discontinuation from opioid use disorder treatment: retrospective observational study	Develop and evaluate a machine learning based two-stage clinical decision-making framework for predicting which patients will discontinue opioid use disorder treatment within less than a year	USA	Retrospective observational	5190; 2025 (39%)	Treatment discontinuation (at least 2 consecutive months without prescription for buprenorphine)

Hayes et al. (2024) [33]	Development and validation of machine-learning algorithms predicting retention, overdoses, and all-cause mortality among US military veterans treated with buprenorphine for opioid use disorder	Develop and validate machine-learning algorithms predicting retention, overdoses, and all-cause mortality among US military veterans initiating B-MOUD	USA	Retrospective cohort	56547 ; 3986 (7%)	Retention on B-MOUD (continuous coverage with B-MOUD for at least 180 days after treatment initiation without a 30-day gap in treatment)
Hong et al. (2024) [34]	Improving treatment completion for young adults with substance use disorder: Machine learning-based prediction algorithm	Investigate the factors associated with treatment completion among young adults diagnosed with SUD, with a focus on criminal justice system involvement and person-level psycho-social factors	USA	Retrospective	2909; 1489 (51.2 %)	Treatment completion

Liu et al. (2024) [35]	Optimizing methadone dose adjustment in patients with opioid use disorder	Examine whether treatment adherence, urine opiate test results, and the likelihood of treatment discontinuation are valuable metrics for determining methadone dose Develop a model for optimizing the methadone dose adjustment, which may be applied to opioid agonist treatment	Taiwan	Retrospective	71; 7 (9.9%)	Likelihood of treatment continuation and discontinuation
Luo et al. (2024) [36]	Individual-level risk prediction of return to use during opioid use disorder treatment	Develop an individual-level prediction tool for risk of return to use in opioid use disorder	USA	Secondary analysis of randomized controlled trials	2199; 728 (33.1%)	Return to use (4 consecutive weeks of urine drug screen either missing or positive for non-prescribed opioids by week 12 of treatment)

Nasir et al. (2021) [37]	Machine learning-based outcome prediction and novel hypotheses generation for substance use disorder treatment	Propose a novel data analytics approach using machine learning models to discover interaction effects that might be neglected by traditional hypothesis-generating approaches	USA	Retrospective cohort	226940; NR	Success (completed treatment) Dropped out Terminated/arrested (treatment terminated by facility due to patient arrest or noncompliance) Transferred (transferred to another SUD program)
Omranian et al. (2023) [38]	Predicting patient satisfaction with medications for treating opioid use disorder: case study applying natural language processing to reviews of methadone and buprenorphine/naloxone on	Use natural language processing methods on text posted on health-related social media to detect patients' satisfaction with two well-studied OUD medications:	USA	Retrospective cohort	4048; NR	Patient satisfaction with treatment or medication

	health-related social media	methadone and buprenorphine/naloxone				
Shikalgar et al. (2024) [39]	Self-help groups and opioid use disorder treatment: An investigation using a machine learning-assisted robust causal inference framework	Investigate the impact of participation in self-help groups on treatment completion among individuals undergoing medication for opioid use disorder treatment	USA	Cross-sectional	157885; 63329 (40.1%)	Treatment completion
Singh et al. (2019) [40]	A machine learning-based approach for collaborative non-adherence detection during opioid abuse surveillance using a wearable biosensor	Leverage accelerometer and blood volume pulse measurements from a wearable biosensor and use machine-learning for the novel problem of collaborative non-adherence detection in opioid surveillance	USA	Cohort	11; 2 (18.2%)	Collaborative non-adherence (giving one's biosensor to someone else when surveillance is ongoing)

Stafford et al. (2022) [41]	Identifying key risk factors for premature discontinuation of opioid use disorder treatment in the United States: A predictive modeling study	Identify key risk factors for premature exit from opioid use disorder treatment	USA	Cross-sectional	2381902; NR	Premature exit from opioid use disorder treatment
Steele et al. (2018) [42]	Machine learning of functional magnetic resonance imaging network connectivity predicts substance abuse treatment completion	Investigate whether functional network connectivity measures were predictive of substance abuse treatment completion using machine learning pattern classification of functional magnetic resonance imaging data	USA	Prospective cohort	139; 89 (64%)	Treatment completion

Victor et al. (2023) [43]	Applied machine learning analysis: Factors correlated with injection drug use and post-prison medication for opioid use disorder treatment engagement	Explore (1) factors that are correlated with injection drug use compared to non-injection drug use 12 months prior to incarceration; and (2) factors that are correlated with MOUD treatment engagement 1 month post release from prison	USA	Retrospective cohort	40; NR	Treatment engagement
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Table 2. Summary of AI/ML models utilized

Author (Year of Publication)	Type of AI/ML Model(s) Used	Best Reported Model	Input Data Source	Training Dataset, Testing Dataset; Performance Metrics	Top 5 Predictive/Modeling Variables in Best Reported Model	Evaluation Metrics of Best Reported Model
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Afshar et al. (2024) [22]	Tian-EN X-BART Uplift Random Forest	Tian-EN	X-BOT clinical trial's public data repository maintained by the National Institute on Drug Abuse Clinical Trials Network	285 (50%); 285 (50%); 5-fold cross-validation	Homelessness, cocaine use disorder, opioid withdrawal score	Qini value = 4.45 (95% CI: 1.02–7.83) c-for-benefit = 0.63 (95% CI: 0.53–0.68)
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Al Faysal et al. (2024) [23]	Logistic Regression Decision Tree Random Forest Extreme Gradient Boosting Adaboost Random Forest-Extreme Gradient Boosting Ensemble	Random Forest	MarketScan commercial claims data from 2018 to 2021	24298 (80%); 6075 (20%); NR	Panel A (baseline predictors) Age, initial days supply, geographic location, opioid analgesics, chronic pain Panel B (baseline predictors + 1-month proportion of days covered) 1-month proportion of days covered, initial days supply, age, geographic location, chronic pain Panel C (baseline predictors + 3-month proportion of days covered)	Panel A (baseline predictors) C-statistic = 0.76 AUC = 0.39 Panel B (baseline predictors + 1-month proportion of days covered) C-statistic = 0.79 AUC = 0.47 Panel C (baseline predictors + 3-month proportion of days covered) C-statistic = 0.82 AUC = 0.47
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					3-month proportion of days covered, initial days supply, age, chronic pain, geographic location	
Baird et al. (2022) [24]	Random Forest Extreme Gradient Boosting Neural Network Logistic Regression	Random Forest	Nationally representative TEDS-D datasets from SAMHSA, which includes data for admissions and discharges from substance use disorder treatment programs	454635 (70%); 194844 (30%); NR	Type of service discharged from, frequency of use of primary substance at discharge, DSM diagnosis, length of stay, age	AUC = 0.881 Accuracy = 80% Precision = 65.5% Recall = 83% F-1 = 73.2%

Baird et al. (2023) [25]	Logistic Regression Gradient Boosting Machine Random Forest Deep Learning	Random Forest	De-identified census data from a substance use disorder outpatient treatment program from 2018 to 2021, including electronic records, paper records, and chart reviews	179 (70%); 77 (30%); 10-fold cross-validation	Number of self-help groups attended during treatment, number of goals set by client, referral source, secondary drug of choice, age	Length-of-stay (>90 days) AUC = 0.782, Accuracy = 0.747 Successful treatment completion AUC = 0.886, Accuracy = 0.831
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Bharat et al. (2021) [26]	Deep Learning	Deep Learning	Data from Opioid Agonist Treatment Safety study derived from Electronic Recording and Reporting of Controlled Drugs system	NR; 5 x 2 cross validation	Predictors of poor retention Dosing point (private), medication type (buprenorphine), incarcerated in the year prior to opioid agonist treatment entry, remoteness (city), personal history of medical treatment noncompliance Predictors of improved retention First entry in 2015-2017 vs 2006-2008, first entry in 2012-2014 vs 2006-2008, mixed anxiety and	R-squared = 0.075 (95% CI: 0.074–0.077) D statistic = 0.584 (95% CI: 0.577–0.590) Harrell's C = 0.610 (95% CI: 0.609–0.611) Integrated Brier score = 0.220 (95% CI: 0.219–0.220)
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					depressive disorder, abdominal pain, other and unspecified abdominal pain	
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Burgess-Hull et al. (2023) [27]	Logistic Regression Logistic Ridge Regression Extreme Gradient Boosting	Extreme Gradient Boosting	Urine drug screen and attendance records, stored in electronic health record systems	NR; 10-fold grouped nested cross validation	Treatment adherence at current appointment, treatment adherence rate, urine positive for fentanyl, cumulative sum of treatment adherent appointments , lagged urine positive for fentanyl	AUROC = 0.87 (95% CI: 0.87–0.88) AUPR = 0.95 Threshold: 0.5 Accuracy = 0.88 Sensitivity = 0.96 Specificity = 0.57 PPV = 0.9 NPV = 0.76 F-1 = 0.93 Threshold: maximum informedness Accuracy = 0.84
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						Sensitivity = 0.86 Specificity = 0.75 PPV = 0.93 NPV = 0.57 F-1 = 0.89
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Cox et al. (2020) [28]	Linear Least Absolute Shrinkage and Selection Operator	Support Vector Machine	Semi-structured assessment for drug dependence and alcoholism for a cohort of patients recruited for genetic studies of opioid, cocaine and alcohol dependence between 2000 and 2017	NR; NR	African Americans Model 1	African Americans
	Linear Support Vector Machine				Time since last cocaine use, time since last cocaine injection, current age, number of years using heroin, time since weekly cocaine use for one month	Accuracy = 75.4%
	Random Forest					European Ancestry
	Deep Neural Network					Accuracy = 79.4%
					African Americans Model 2	
					Time since last cocaine use, time since weekly cocaine use for one month, recently had one or more cocaine symptoms, cocaine use >11 times	

					last year, current age	
					African Americans Model 3	
					Current age, number of biological children, currently unemployed, age at maximum weight, number of months employed last year	
					European Ancestry Model 1	
					Time since last cocaine use, time since last cocaine injection, cocaine use >11 times last year, age at maximum	

					weight, current age	
					European Ancestry	
					Model 2	
					Time since last cocaine use, time since last cocaine injection, cocaine use >11 times last year, age at maximum weight, current age	
					European Ancestry	
					Model 3	
					Current age, age at maximum weight, age when last had antisocial behaviours, body mass index, current weight	

Eddie et al. (2024) [29]	Random Forest	Random Forest	Global Appraisal of Individual Needs dataset and United States Census Bureau data	11905 (75%); 3968 (25%); NR	Low employment, low alcohol sales, high poverty, enrolled in school, high population density	AUC = 0.71 Accuracy = 0.75 Sensitivity = 0.91 Specificity = 0.30
Gottlieb et al. (2022) [30]	Logistic Regression Support Vector Machine Random Forest AdaBoost Gentle Boost LogitBoost Robust Boosting Totally Corrective Boosting	Random Forest	Self-reported questionnaires and health evaluation by nurse practitioners or physician assistants	NR; 10-fold cross validation	Overdose in the past, improvement in quality of life, relapse since joining the program	90 days Sensitivity = 0.81 ± 0.02 Specificity = 0.65 ± 0.05 120 days Sensitivity = 0.86 ± 0.03 Specificity = 0.66 ± 0.02

Haredasht et al. (2024) [31]	Logistic Regression Random Forest Extreme Gradient Boosting	Random Forest	Electronic health record data. from Stanford University's healthcare system and Holmusk's NeuroBlu database	NR; 5-fold cross-validation	NR	Stanford External Precision = 84.1 ± 1.1 Recall = 78.3 ± 0.7 Stanford Internal Precision = 73.61 ± 1.6 Recall = 71.17 ± 2.5 Neuroblu External Precision = 62.9 ± 0.5 Recall = 88.5 ± 1.0 Neuroblu Internal Precision = 87.3 ± 0.6 Recall = 77.3 ± 1.2
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Hasan et al. (2021) [32]	Logistic Regression Decision Tree Random Forest Extreme Gradient Boosting Support Vector Machine Artificial Neural Network	Random Forest	Administrative data from the Massachusetts All Payer Claims Database	NR; 5-fold cross-validation	NR	First stage models with baseline predictors Precision = 0.56 ± 0.02 Recall = 0.49 ± 0.02 C-statistic = 0.59 ± 0.02 F-1 score = 0.53 ± 0.02 Second stage models with 2 months PDC and baseline predictors Precision = 0.68 ± 0.01 Recall = 0.55 ± 0.03 C-statistic = 0.71 ± 0.01 F-1 score = 0.61 ± 0.02
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						Second stage models with 3 months PDC and baseline predictors Precision = 0.71 ± 0.02 Recall = 0.60 ± 0.03 C-statistic = 0.75 ± 0.03 F-1 score = 0.65 ± 0.03
Hayes et al. (2024) [33]	Multiple Logistic Regression Least Absolute Shrinkage and Selection Operator-Type Regression Random Forest Gradient Boosting Machine	Gradient Boosting Machine	Inpatient, outpatient, demographic, and outpatient pharmacy files from the VHA Corporate Data Warehouse	45238 (80%); 11309 (20%); NR	NR	AUC-ROC = 0.72

	Deep Neural Network					
Hong et al. (2024) [34]	Chi-square Automatic Interaction Detection Decision Tree	Chi-square Automatic Interaction Detection Decision Tree	Adult Needs and Strength Assessment and National Outcome Measures data	2618 (90%); 291 (10%); NR	Level of improvement in total actionable items was the strongest predictor Other important predictors mentioned include age, rural/urban living, and criminal justice involvement; however, the contribution of these compared to other variables in the context of the model's predictive performance is unclear	Training dataset AUC = 0.80 Testing dataset AUC = 0.82

Liu et al. (2024) [35]	Neural Network	Neural Network	Daily dosing records and urine opiate test results from patients with opioid use disorder at Taichung Veterans General Hospital	NR; 10-fold cross validation	NR	Mean correlation coefficient = 0.995 ± 0.003 (p<0.001)
Luo et al. (2024) [36]	Least Absolute Shrinkage and Selection Operator Logistic Regression	Logistic Regression	Data from 3 multicenter, pragmatic, randomized clinical trials within the National Institute on Drug Abuse Clinical Trials Network	1650 (75%); 549 (25%); NR	Heroin use days, morphine positive urine drug screen, cocaine positive urine drug screen, urinary drug screen in the first 3 treatment weeks	Treatment entry variables AUROC = 0.68 (95% CI: 0.63–0.72) Opioid use in first 3 weeks AUROC = 0.82 (95% CI: 0.78–0.85) Treatment entry

						variables + opioid use in first 3 weeks AUROC = 0.82 (95% CI: 0.78–0.85)
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Nasir et al. (2021) [37]	Random Forest Artificial Neural Network Extreme Gradient Boosting	Extreme Gradient Boosting	SAMHSA's Treatment Episode Data Set — Discharges (TEDS-D) for 2015-2017 and the FBI's Uniform Crime Reporting Program Dataset	113470 (50%); 113470 (50%); 5-fold cross-validation	Unclear except length of stay is the strongest predictor	Success AUC = 0.8931 Balanced accuracy = 0.7743 Precision = 0.7789 Recall = 0.9169 Dropout AUC = 0.8663 Balanced accuracy = 0.7351 Precision = 0.6418 Recall = 0.5695 Arrest/terminate AUC = 0.8225
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						Balanced accuracy = 0.5624 Precision = 0.4566 Recall = 0.1385 Transferred AUC = 0.8829 Balanced accuracy = 0.6684 Precision = 0.4984 Recall = 0.3791
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Omranian et al. (2023) [38]	Logistic regression Elastic Net Least absolute Shrinkage and Selection Operator Random Forest Ridge Regression Extreme Gradient Boosting	Ridge Regression	Two health care forums, Drugs.com and WebMD containing self-reported experiences of medications	NR; 10-fold cross-validation	NR	AUC = 78.1 F-1 Score = 90.8%
Shikalgar et al. (2024) [39]	Lasso Regression Decision Trees Random Forest Extreme Gradient Boosting	Random Forest	Health data records from the SAMHSA's public data repository (TEDS-D)	110519 (70%); 47366 (30%); 10-fold cross-validation	Self-help, service setting at admission, census division, frequency of use at admission, treatment referral source	AUC = 0.76 Precision = 0.90 Specificity = 0.93 Recall = 0.40

Singh et al. (2019) [40]	Random Forest AdaBoost Support Vector Machine Logistic Regression	Random Forest	Accelerometer and blood volume pulse measurements from a wearable biosensor	NR; 5-fold cross-validation	NR	AUC = 0.94 Accuracy = 90.96%
Stafford et al. (2022) [41]	Random Forest Classification and Regression Tree Bagged Classification and Regression Tree Boosted Classification and Regression Tree	Random Forest	OUD treatment episodes in the Treatment Episode Data Set-Discharges (TEDS-D)	1100838 (75%); 366945 (25%); 10-fold cross-validation	Service setting, geographic region, primary source of payment, source of referral, primary income source	ROC-AUC = 74%

Steele et al. (2018) [42]	Support Vector Machine (functional network connectivity measures alone, clinical assessments alone, and functional network connectivity measures + clinical assessments)	Support Vector Machine (functional connectivity measures + clinical assessments)	Assessments including Psychopathy Checklist-Revised, Vocabulary and Matrix Reasoning subtests of the Wechsler Adult Intelligence Scale, self-report measures of anxiety, depression, motivation for change and Addiction Severity Index, and fMRI data	125 (90%); 14 (10%); 2, 5 and 10-fold cross-validation	NR	Accuracy = 81.29% Sensitivity = 82.24% Specificity = 78.13% PPV = 92.63% NPV = 56.82%
Victor et al. (2023) [43]	Chi-square Automatic Interaction Detection Decision Tree	Chi-square Automatic Interaction Detection Decision Tree	Data from a Midwestern re-entry program for incarcerated individuals with co-occurring opioid use and a mental	NR; NR	Psychiatric medication use, intravenous drug use, participation in substance use treatment	NR

			health disorder survey			
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Table 3. Number of included studies per year

Year of Publication	Number of Included Studies, n
2018	1
2019	1
2020	1
2021	3
2022	3
2023	4
2024	9

Table 4. Number of included studies per country

County of Publication	Number of Included Studies, n
Australia	1
Taiwan	1
USA	20

Table 5. Number of included studies per treatment adherence outcome

Adherence Outcome	Number of Included Studies, n
Return to use	1
Treatment induction	1
Treatment engagement and adherence	3
Treatment retention	4

Treatment satisfaction	1
Treatment completion	6
Opioid cessation	1
Treatment non-adherence	2
Treatment discontinuation, cessation or drop out	7

Table 6. Number of included studies per best reported model

Best Reported Model	Number of Included Studies, n
Chi-square Automatic Interaction Detection Decision Tree	2
Deep Learning	1
Extreme Gradient Boosting	2
Gradient Boosting Machine	1

Logistic Regression	1
Neural Network	1
Random Forest	10
Ridge Regression	1
Support Vector Machine	2
Tian-EN	1

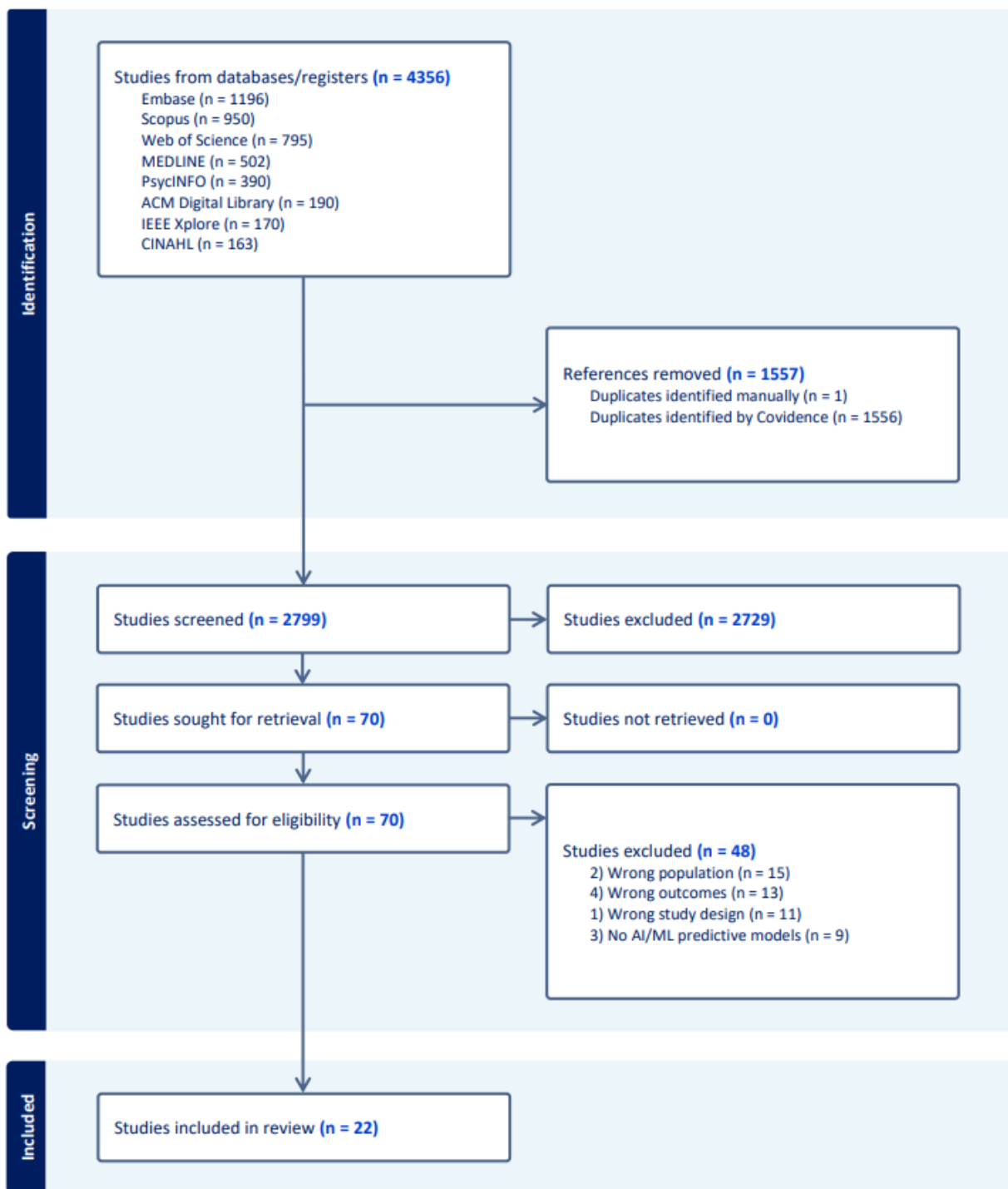


Figure 1: Preferred Reporting Items for Scoping reviews and Meta-Analysis (PRISMA) flow chart for the study

S1. Complete List of Predictive or Modeling Variables for Each Study

Author (Year of Publication)	Predictive/Modeling Variables
Afshar et al. (2024)	Age at randomization, gender, ethnicity, education, looking for work, late randomization, high severity addiction, heroin as the major drug of abuse, any heroin use, heroin use route, cannabis use (30 days), cocaine use (30 days), opiate use onset, opioid withdrawal score, nicotine use onset, primary drug cost per day, living with an alcoholic, living with a drug user, homeless, on parole, smoking, chronic pain for 6 months, moderate or extreme pain, psychiatric diagnosis, moderate or extreme anxiety/depression, Hamilton depression score, alcohol use disorder, amphetamine use disorder, benzodiazepine use disorder, cannabis use disorder, cocaine use disorder previous methadone/ buprenorphine treatment success, strong medication preference, no medication preference, preference for SUB-BUP, no preference for SUB-BUP, no medication preference, preference for ER-NTX, no preference for ER-NTX
Al Faysal et al. (2024)	Age, sex, geographic region, residence, insurance plan type, relationship to policyholder, geographic location, chronic pain condition, HIV/AIDS condition, Hepatitis C condition, depressive disorder, anxiety, post-traumatic stress disorder, bipolar disorder, schizophrenia, non-opioid drug use disorder, alcohol use disorder, opioid use disorder severity level, cannabis use disorder severity level, Charlson Comorbidity Index, number of days in first buprenorphine subscription, use of antidepressants, use of antipsychotics, use of mood stabilizers, use of benzodiazepines, use of non-benzo, use of stimulants, use of opioid analgesics, number of inpatient visits, number of emergency department visits, number of outpatient visits, proportion of days covered measure 1-month of treatment initiation, proportion of days covered measure 3-month of treatment initiation
Baird et al. (2022)	Age, DSM diagnosis, length of stay, primary substance used, secondary substance used, referral source, frequency of use of primary substance at admission, frequency of use of primary substance at discharge, employment status at discharge, type of service the patient is discharged from

Baird et al. (2023)	Age, highest grade completed, number of self help groups attended 30 days prior to treatment, number of self help groups attended during treatment, number of goals set by client, referral source, drug of choice, secondary drug of choice, number of previous treatment episodes, linked to supplemental nutrition assistance program after intake
Bharat et al. (2021)	Age at treatment initiation, gender, Aboriginal and/or Torres Strait Islander, remoteness, incarcerated prior to opioid agonist treatment entry, number of convictions prior to opioid agonist treatment entry, medication type, year of first opioid agonist treatment entry, dosing point, tenure of prescriber's opioid agonist prescribing, prescriber working in multiple locations, history of noncompliance with medical treatment and regimen, outpatient treatment for DUDs, schizophrenia, homelessness, prior treatment for harmful opioid use, tobacco dependence, alcohol use disorder, occupant of vehicle in an accident, mixed anxiety and depressive disorder, abdominal pain, other and unspecified abdominal pain, treatment for opioid dependence, treatment for opioid withdrawal, personal history of tobacco use disorder, current tobacco use, 4-aminophenol derivatives
Burgess-Hull et al. (2023)	BlueCross as primary insurance type, Medicaid as primary insurance type, Medicare as primary insurance type, possible toxicology adulteration, urine positive for fentanyl, urine positive for non-fentanyl opioid, urine positive for cocaine, urine positive for benzodiazepine, urine positive for norbuprenorphine, urine positive for amphetamine, cumulative sum (cocaine, benzodiazepine, amphetamine, gabapentin), treatment adherence rate, attendance rate, treatment adherence rate x number of appointments attended (interaction), treatment adherent appointments, number of appointments attended, lagged urine positive for fentanyl, lagged urine positive for non-fentanyl opioids, lagged urine positive for norbuprenorphine, lagged urine positive for cocaine, lagged urine positive for norbuprenorphine, treatment adherence at current appointment, treatment adherence at last appointment

Cox et al. (2020)	For African Americans Current age, sex, time since first opioid treatment, older age at first opioid treatment, number of years using heroin, depressed after reducing cocaine use, time since last injected cocaine, time since last used cocaine, time since last stayed high in cocaine, used cocaine <11 times within year of interview, treated in outpatient program for cocaine use, time since first cocaine craving, never injected cocaine, often used marijuana more than intended to, mixed alcohol and drugs > 3 times in 12 months, time since last had alcohol symptoms lasting >1 month, smoked less frequently after waking up, older age at first cigarette, 2 marijuana symptoms lasting a month, HIV positive, previous health always better than current health For Individuals with European Ancestry Current age, household income, body mass index, time since last cocaine use, older age at first heavy opioid use number of years using heroin, time since last cocaine injection, time since last had alcohol symptoms that last >1 month, >20 outpatient visits for drug/psychiatric problems in the last year, time since opioid treatment initiation, used cocaine >11 times in last year, time since first used opioid 1/week for >1 month, have injected cocaine, older age at first heavy cocaine use, marijuana interfered with work/home, time since starting opioid self-help group, used tobacco but not addicted, disclosed problems with cocaine usage to professional, stopped using stimulants for >3 months, drinking resulting in objections or problems with family and work, <3 ASP criteria in 12-month period, often failed to pay debts, suspended or expelled from school, time since last suicidal ideation, age last had antisocial behaviour, no fear or most disturbing/traumatizing event, avoided scene that reminded of traumatic event, OCD behaviours when depressed, feeling distracted, unsafely raced cars, depression always started with drug problems, number of depression symptoms, outstanding emotional problem
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Eddie et al. (2024)	Age, sex, race/ethnicity, employment status, substance use severity, prior substance abuse treatment, treatment motivation index, treatment resistance index, victimization, substance use age of onset >15, current withdrawal symptoms, dual diagnosis, depression symptoms, emotional problem scale, ADHD, conduct disorder, post-traumatic stress disorder, suicide problems, criminal justice involvement, living risk index, social risk index, homelessness experience, lack of peers in recovery, general crime scale, self-efficacy scale, population density, public assistance, income inequality, health care, count of liquor places near treatment center
Gottlieb et al. (2022)	Gender, quality of life improvement, have relapsed since joining the program, have used prescription oxycodone, past legal problems, substance use history of benzodiazepines, veteran status, first time taking medications for opioid use disorder, housing status, prior use of barbiturates, substance use history of hallucinogens, substance use history of non-prescription opiates, substance use history of cocaine, prior use of non-prescription suboxone, substance use history of alcohol, substance use history of marijuana, substance use history of heroin, taken medications for opioid use disorder from inpatient services, prior use of cocaine, prior use of nicotine or tobacco, prior arrest for drug charges, religiosity, prior arrest for parole or probation violations, ever taken medications for opioid use disorder in outpatient services
Haredasht et al. (2024)	Diagnosis of opioid dependence, diagnosis of limb swelling, polyethylene glycol-based laxative, oxycodone extended release, acetaminophen, age at BUP-NAL initiation, 12-lead ECG, diagnosis of opioid withdrawal, diphenhydramine, docusate, aspirin, lidocaine low dose buprenorphine, IV fluids, IV thiamine
Hasan et al. (2021)	Age, income, gender, patient location (urban/rural), insurance type, provider speciality, addiction treatment specialist, pain specialist, early treatment adherence, PDC in first 2 months, PDC in first 3 months

Hayes et al. (2024)	Age, gender, length of time for Veterans Health Administration (VHA) enrollment, religion, race, ethnicity, marital status, VHA priority status, state of residence, US census division of residence, rural-urban commuting area code, miles from closest VHA facility, minutes of drive-time to the closest VHA facility, employment status, VHA disability status, VHA service-connected disability eligibility status, VHA pension eligibility status, smoking status, next of kin, days' supply of initial buprenorphine, dose of initial buprenorphine, days covered of initial buprenorphine, inpatient or outpatient initiation of buprenorphine, days between OUD diagnosis and buprenorphine initiation, year of buprenorphine initiation, month of buprenorphine initiation, methadone orders between OUD diagnosis and buprenorphine initiation, days' supply of methadone between OUD diagnosis and buprenorphine initiation, naltrexone orders between OUD diagnosis and buprenorphine initiation, days' supply of naltrexone between OUD diagnosis and buprenorphine initiation, number of previous buprenorphine episodes, number of previous failed buprenorphine episodes, days covered from previous buprenorphine episodes, total buprenorphine dose from previous buprenorphine episodes, average buprenorphine dose from previous buprenorphine episodes, time since last buprenorphine episode, binary indicator for whether the difference between buprenorphine initiation and start of data is 365 days, fractional value for number of days between buprenorphine initiation and fiscal year 2006, number of days between buprenorphine initiation and start of fiscal year 2006, number of emergency room visits, number of primary care/medicine visits, number of mental health visits, number of pain clinic visits, whether or not brief addiction monitor (BAM) or brief addiction monitor-revised (BAM-R) has been administered, number of BAM or BAM-R administrations, days between buprenorphine initiation and BAM administration, individual constructs within BAM measured over the past 30 days, cannabis use disorder, cocaine use disorder, tobacco use disorder, alcohol use disorder, non-opioid drug use disorder, endocarditis, Hepatitis B, Hepatitis C, HIV, post-traumatic stress disorder, opioid overdose, non-opioid overdose, any mental health disorder, depression, anxiety, bipolar disorder, psychosis, suicide or self-inflicted injury, elixhauser, comorbidity count, opioid prescription duration of action, opioid prescription schedule, average morphine milligram equivalents, total opioid days covered, number of medication fills received, number of unique VHA categories of medications received, number of naloxone prescription received, days covered of antidepressants, days covered of benzodiazepines, days covered of skeletal muscle relaxants, drugs covered of non-opioid analgesics, days covered of hypnotics, Social Vulnerability Index (SVI) composite score, SVI household composition & disability score, SVI housing type & transportation score, SVI socioeconomic score, homelessness, justice-involvement, provider speciality, provider credentials, VHA or non-VHA
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	prescriber, trainee, number of buprenorphine written by prescriber, number of methadone prescriptions written by prescriber, number of naltrexone prescriptions written by prescriber, number of patients prescriber wrote buprenorphine for, number of patients prescriber wrote methadone for, number of patients prescriber wrote naltrexone for, percent of patients retained on buprenorphine in the year by prescriber, whether veteran received buprenorphine from closest VHA facility, facility type, whether facility had an inpatient detoxification program, whether facility had a methadone clinic, whether facility had a substance abuse disorder residential treatment program, flag for VHA station
Hong et al. (2024)	Gender, race/ethnicity, types of substances used, opioid use, age of first use for the primary substance, Adult Needs and Strength Assessment strengths, total actionable items, rural/urban residence, living arrangement, employment, education, any justice system involvement
Liu et al. (2024)	HIV positivity, referral by the criminal justice system, 90-day urine opiate test results, last methadone dosage taken, willingness to receive treatment, 7-day treatment adherence, 30-day treatment adherence
Luo et al. (2024)	Heroin use days, morphine-positive urine drug screen, cocaine-positive urine drug screen, heroin injection in the past 30 days
Nasir et al. (2021)	Age, individual as the referral source, drug use care provider as referral source, health care provider as referral source, school as referral source, employer as referral source, community referral, education, primary substance frequency at admission, secondary substance frequency at admission, homeless, dependent living arrangements, primary substance frequency, length of stay, treatment locality population, treatment locality total crime cases, treatment locality disorderly conduct cases, change in self-help attendance, short-term rehab services, change in arrests, planned medication-assisted opioid therapy, treatment locality vagrancy cases, full-time employment status, substance use type (alcohol only), CBSA 35300, Services (short term rehab) CBSA 31700, CBSA 17200, CBSA 31700, CBSA 45860

Omranian et al. (2023)	Symptoms (suffocate, breathing problem, dehydration, vomit, confusion, restlessness, disoriented, muscle weakness, mood changes, depressing), drugs (hydroxyzine, luvox, temazepam, vitamins, miralax, fioricet, Narcan, magnesium citrate, Adderall, Ambien), illness (adrenal crisis, delirium, reflex sympathetic dystrophy, cyst, hydrocephalus, chronic headaches, sciatica, fibromyalgia, herniated discs, degenerative joint disease), duration of treatment, oral sensation, side effects, insurance, doctor visit
Shikalgar et al. (2024)	Age at first use, age at admission, gender, ethnicity, education, employment status, marital status, service setting at admission, census division, year of discharge from substance use treatment, treatment referral source, frequency of drug use at admission, primary substance route of use, second substance use at admission, injecting substance as route of administration, no prior treatment episodes, cocaine use, heroin use, marijuana use, alcohol use, benzodiazepine use, methamphetamine use, hallucinogen use, arrests 30 days prior to discharge, comorbid psychological problems, homeless, veteran status
Singh et al. (2019)	Features taken from accelerometer Mean, medica, skewness, variance, standard deviation, mean crossing rate, maximum value, minimum value, mean derivatives, interquartile range, zero crossing rate, kurtosis Features taken from blood volume pulse Number of peaks, mean of peaks, standard deviation of peaks, root mean square, distance from maximum to minimum of peaks, mean peak-to-peak distance, standard deviation of peak-to-peak distance, power spectral density of signal

Stafford et al. (2022)	Age, sex race, ethnicity, marital status, education, employment status, primary income source, service setting, geographic region, primary source of payment, referral source, planned use of medication for OUD, maximum frequency of use, days waited to enter treatment, prior treatment episodes, number of substances used, age of first substance use, infection use, heroin use, stimulant use, alcohol use, marijuana use, hallucinogen use, sedative use, inhalant use, tranquilizer use, arrests in 30 days prior to treatment, living situation, veteran status, diagnosed psychological problem
Steele et al. (2018)	Age, sex, IQ, years of drug use, psychopathy, anxiety, depressive symptomatology, functional network connectivity between networks incorporating the anterior cingulate cortex; striatum; and insula
Victor et al. 2023	Psychiatric medication use, intravenous drug use, participation in substance use treatment

S2. Complete Search Strategy

Medline (Ovid)	502
1. exp Artificial Intelligence/ OR exp Decision Trees/ OR exp Data Mining/ 2. (Artificial Intelligence OR neural network* OR Support Vector Machine* OR machine learning OR Deep Learning OR Decision Tree* OR Data Mining OR text mining OR Naive bayes OR k-nearest neighbor* OR Random Forest OR gradient boosting OR adaptive boosting OR regression tree* OR Natural Language Processing OR Automated reasoning OR Predictive analytic* OR supervised learning OR unsupervised learning).mp. 3. ((Prediction model* OR predictive model*) adj4 (algorithm or AI or ML or DL or logistic regression)).mp.	
4. (exp Morphinans/ OR exp Narcotics/ OR exp Substance-Related Disorders/ OR exp Opiate Alkaloids/ OR exp Analgesics, Opioid/ OR exp Opium/ OR exp Opiate Substitution Treatment/) 5. (Opioid* OR Opiat* OR Avinza* OR Buprenorphine* OR Codeine* OR Fentanyl* OR Heroin* OR Hydrocodone* OR Kadian* OR Methadone* OR Morphine* OR Naltrexone* OR Narcot* OR naloxone* OR Opana* OR Opium* OR Oxycodone* OR Oxymorphone* OR OxyContin* OR Percocet* OR Suboxone* OR Subutex OR Vivitrol* OR Vicodin* OR Substance use* OR substance dependen* OR Substance disorder* OR Substance abuse* OR Substance misuse*).mp.	

[illegible]

Embase Classic	1196
1. exp “artificial intelligence”/ OR exp “machine learning”/ OR exp “Natural Language Processing”/ OR exp “Decision Tree”/ 2. (“artificial intelligence” OR “neural network*” OR “Support Vector Machine*” OR “machine learning” OR “Deep Learning” OR “Decision Tree*” OR “Data Mining” OR “text mining” OR “Naive bayes” OR “k-nearest neighbor*” OR “random forest*” OR “gradient boosting” OR “regression tree*” OR “adaptive boosting” OR “Natural Language Processing” OR “Automated reasoning” OR “Predictive analytic*” OR “supervised learning” OR “unsupervised learning”).mp. 3. ((“Prediction model*” OR “predictive model*”) ADJ4 (algorithm OR AI OR ML OR DL OR “logistic regression”)).mp.	
4. (exp morphinan derivative/ OR exp narcotic agent/ OR exp drug dependence/ OR exp opiate/ OR exp opiate substitution treatment/ OR exp opiate antagonist/) 5. (Opioid* OR Opiat* OR Avinza* OR Buprenorphine* OR Codeine* OR Fentanyl* OR Heroin* OR Hydrocodone* OR Kadian* OR Methadone* OR Morphine* OR Naltrexone* OR Narcot* OR naloxone* OR Opana* OR Opium* OR Oxycodone* OR Oxymorphone* OR OxyContin* OR Percocet* OR Suboxone* OR Subutex OR Vivitrol* OR Vicodin* OR Substance use* OR substance dependen* OR Substance disorder* OR Substance abuse* OR Substance misuse*).mp.	

1. exp "artificial intelligence"/ OR exp "Neural Networks"/ OR exp "Data Mining"/
2. ("artificial intelligence" OR "neural network*" OR "Support Vector Machine*" OR "machine learning" OR "Deep Learning" OR "Decision Tree*" OR "Data Mining" OR "text mining" OR "Naive bayes" OR "k-nearest neighbor*" OR "random forest*" OR "gradient boosting" OR "regression tree*" OR "Natural Language Processing" OR "adaptive boosting" OR "Automated reasoning" OR "Predictive analytic*" OR "supervised learning" OR "unsupervised learning").mp.
3. (("Prediction model*" OR "predictive model*") ADJ4 (algorithm OR AI OR ML OR DL OR "logistic regression")).mp.

4. (exp Opiates/ OR exp Narcotic Drugs/ OR exp "substance related and addictive disorders"/ OR exp narcotic agonists/ OR exp drug dependency/ OR exp analgesic drugs/ OR exp opioid epidemic/ OR exp "substance use treatment"/)
5. (Opioid* OR Opiat* OR Avinza* OR Buprenorphine* OR Codeine* OR Fentanyl* OR Heroin* OR Hydrocodone* OR Kadian* OR Methadone* OR Morphine* OR Naltrexone* OR Narcot* OR naloxone* OR Opana* OR Opium* OR Oxycodone* OR Oxymorphone* OR OxyContin* OR Percocet* OR Suboxone* OR Subutex OR Vivitrol* OR Vicodin* OR Substance use* OR substance dependen* OR Substance disorder* OR Substance abuse* OR Substance misuse*).mp.

((TITLE-ABS-KEY("artificial intelligence")) OR (TITLE-ABS-KEY("neural network*")) OR (TITLE-ABS-KEY("Support Vector Machine*")) OR (TITLE-ABS-KEY("machine learning")) OR (TITLE-ABS-KEY("Deep Learning")) OR (TITLE-ABS-KEY("Decision Tree*")) OR (TITLE-ABS-KEY(("Prediction model*" OR "predictive model*") W/4 (algorithm OR AI OR ML OR DL OR "logistic regression")) OR (TITLE-ABS-KEY("Data Mining")) OR (TITLE-ABS-KEY("text mining")) OR (TITLE-ABS-KEY("Naive bayes")) OR (TITLE-ABS-KEY("k-nearest neighbo*")) OR (TITLE-ABS-KEY("gradient boosting")) OR (TITLE-ABS-KEY("regression tree*")) OR (TITLE-ABS-KEY("adaptive boosting")) OR (TITLE-ABS-KEY("Natural Language Processing")) OR (TITLE-ABS-KEY("Automated reasoning")) OR (TITLE-ABS-KEY("Predictive analytic*")) OR (TITLE-ABS-KEY("supervised learning")) OR (TITLE-ABS-KEY("unsupervised learning")))

((TITLE-ABS-KEY(Morphin*)) OR (TITLE-ABS-KEY("Substance-Related Disorder*")) OR (TITLE-ABS-KEY(Opioid*)) OR (TITLE-ABS-KEY(Opiat*)) OR (TITLE-ABS-KEY(Avinza*)) OR (TITLE-ABS-KEY(Buprenorphine*)) OR (TITLE-ABS-KEY(Codeine*)) OR (TITLE-ABS-KEY(Fentanyl*)) OR (TITLE-ABS-KEY(Heroin*)) OR (TITLE-ABS-KEY(Hydrocodone*)) OR (TITLE-ABS-KEY(Kadian*)) OR (TITLE-ABS-KEY(Methadone*)) OR (TITLE-ABS-KEY(Morphine*)) OR (TITLE-ABS-KEY(Naltrexone*)) OR (TITLE-ABS-KEY(Narcot*)) OR (TITLE-ABS-KEY(naloxone*)) OR (TITLE-ABS-KEY(Opana*)) OR (TITLE-ABS-KEY(Opium*)) OR (TITLE-ABS-KEY(Oxycodone*)) OR (TITLE-ABS-KEY(Oxymorphone*)) OR (TITLE-ABS-KEY(OxyContin*)) OR (TITLE-ABS-KEY(Percocet*)) OR (TITLE-ABS-KEY(Suboxone*)) OR (TITLE-ABS-KEY(Subutex)) OR (TITLE-ABS-KEY(Vivitrol*)) OR (TITLE-ABS-KEY(Vicodin*)) OR (TITLE-ABS-KEY("Substance use*")) OR (TITLE-ABS-KEY("substance dependen*")) OR (TITLE-ABS-KEY("Substance disorder*")) OR (TITLE-ABS-KEY("Substance abuse*")) OR (TITLE-ABS-KEY("Substance misuse*")))

((TITLE-ABS-KEY(Adher*)) OR (TITLE-ABS-KEY(nonadher*)) OR
(TITLE-ABS-KEY(non-adher*)) OR (TITLE-ABS-KEY(non adher*)) OR
(TITLE-ABS-KEY(discontin*)) OR (TITLE-ABS-KEY(complian*)) OR
(TITLE-ABS-KEY(Noncomplian*)) OR (TITLE-ABS-KEY(non-complian*)) OR
(TITLE-ABS-KEY(non complian*)) OR (TITLE-ABS-KEY(nonappear*)) OR
(TITLE-ABS-KEY(non-appear*)) OR (TITLE-ABS-KEY(dropout*)) OR
(TITLE-ABS-KEY(drop-out*)) OR (TITLE-ABS-KEY(non-attend*)) OR
(TITLE-ABS-KEY(nonattend*)) OR (TITLE-ABS-KEY(attend*)) OR (TITLE-ABS-KEY(noshow*))
OR (TITLE-ABS-KEY(no-show*)) OR (TITLE-ABS-KEY(absent*)) OR
(TITLE-ABS-KEY(absence*)) OR (TITLE-ABS-KEY(retention*)) OR (TITLE-ABS-KEY(retain*))
OR (TITLE-ABS-KEY(attrition*)) OR (TITLE-ABS-KEY(contin*)) OR
(TITLE-ABS-KEY(follow-through*)) OR (TITLE-ABS-KEY(persist*)) OR
(TITLE-ABS-KEY(engagement*)) OR (TITLE-ABS-KEY(refusal*)) OR
(TITLE-ABS-KEY(non-treatment*)) OR (TITLE-ABS-KEY(non treatment*)) OR
(TITLE-ABS-KEY(nontreatment*)) OR (TITLE-ABS-KEY(“return to use”)))

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(( TITLE-ABS-KEY ( "artificial intelligence" ) ) OR ( TITLE-ABS-KEY ( "neural network*" ) ) OR (
TITLE-ABS-KEY ( "Support Vector Machine*" ) ) OR ( TITLE-ABS-KEY ( "machine learning" ) )
OR ( TITLE-ABS-KEY ( "Deep Learning" ) ) OR ( TITLE-ABS-KEY ( "Decision Tree*" ) ) OR (
TITLE-ABS-KEY ( ( "Prediction model*" OR "predictive model*" ) W/4 ( algorithm OR ai OR ml OR
dl OR "logistic regression" ) ) ) OR ( TITLE-ABS-KEY ( "Data Mining" ) ) OR ( TITLE-ABS-KEY (
"text mining" ) ) OR ( TITLE-ABS-KEY ( "Naive bayes" ) ) OR ( TITLE-ABS-KEY ( "k-nearest
neighbo*" ) ) OR ( TITLE-ABS-KEY ( "gradient boosting" ) ) OR ( TITLE-ABS-KEY ( "regression
tree*" ) ) OR ( TITLE-ABS-KEY ( "adaptive boosting" ) ) OR ( TITLE-ABS-KEY ( "Natural
Language Processing" ) ) OR ( TITLE-ABS-KEY ( "Automated reasoning" ) ) OR ( TITLE-ABS-KEY
( "Predictive analytic*" ) ) OR ( TITLE-ABS-KEY ( "supervised learning" ) ) OR ( TITLE-ABS-KEY
( "unsupervised learning" ) ) ) AND ( ( TITLE-ABS-KEY ( morphin* ) ) OR ( TITLE-ABS-KEY (
"Substance-Related Disorder*" ) ) OR ( TITLE-ABS-KEY ( opioid* ) ) OR ( TITLE-ABS-KEY (
opiat* ) ) OR ( TITLE-ABS-KEY ( avinza* ) ) OR ( TITLE-ABS-KEY ( buprenorphine* ) ) OR (
TITLE-ABS-KEY ( codeine* ) ) OR ( TITLE-ABS-KEY ( fentanyl* ) ) OR ( TITLE-ABS-KEY (
heroin* ) ) OR ( TITLE-ABS-KEY ( hydrocodone* ) ) OR ( TITLE-ABS-KEY ( kadian* ) ) OR (
TITLE-ABS-KEY ( methadone* ) ) OR ( TITLE-ABS-KEY ( morphine* ) ) OR ( TITLE-ABS-KEY (
naltrexone* ) ) OR ( TITLE-ABS-KEY ( narcot* ) ) OR ( TITLE-ABS-KEY ( naloxone* ) ) OR (
TITLE-ABS-KEY ( opana* ) ) OR ( TITLE-ABS-KEY ( opium* ) ) OR ( TITLE-ABS-KEY (
oxycodone* ) ) OR ( TITLE-ABS-KEY ( oxymorphone* ) ) OR ( TITLE-ABS-KEY ( oxycontin* ) )
OR ( TITLE-ABS-KEY ( percocet* ) ) OR ( TITLE-ABS-KEY ( suboxone* ) ) OR (
TITLE-ABS-KEY ( subutex ) ) OR ( TITLE-ABS-KEY ( vivitrol* ) ) OR ( TITLE-ABS-KEY (
vicodin* ) ) OR ( TITLE-ABS-KEY ( "Substance use*" ) ) OR ( TITLE-ABS-KEY ( "substance
dependen*" ) ) OR ( TITLE-ABS-KEY ( "Substance disorder*" ) ) OR ( TITLE-ABS-KEY (
"Substance abuse*" ) ) OR ( TITLE-ABS-KEY ( "Substance misuse*" ) ) ) AND ( ( TITLE-ABS-KEY
( adher* ) ) OR ( TITLE-ABS-KEY ( nonadher* ) ) OR ( TITLE-ABS-KEY ( non-adher* ) ) OR (
TITLE-ABS-KEY ( non AND adher* ) ) OR ( TITLE-ABS-KEY ( discontinu* ) ) OR (
TITLE-ABS-KEY ( complian* ) ) OR ( TITLE-ABS-KEY ( noncomplian* ) ) OR ( TITLE-ABS-KEY
( non-complian* ) ) OR ( TITLE-ABS-KEY ( non AND complian* ) ) OR ( TITLE-ABS-KEY (
nonappear* ) ) OR ( TITLE-ABS-KEY ( non-appear* ) ) OR ( TITLE-ABS-KEY ( dropout* ) ) OR (
TITLE-ABS-KEY ( drop-out* ) ) OR ( TITLE-ABS-KEY ( non-attend* ) ) OR ( TITLE-ABS-KEY (
nonattend* ) ) OR ( TITLE-ABS-KEY ( attend* ) ) OR ( TITLE-ABS-KEY ( noshow* ) ) OR (
TITLE-ABS-KEY ( no-show* ) ) OR ( TITLE-ABS-KEY ( absent* ) ) OR ( TITLE-ABS-KEY (
absence* ) ) OR ( TITLE-ABS-KEY ( retention* ) ) OR ( TITLE-ABS-KEY ( retain* ) ) OR (
TITLE-ABS-KEY ( attrition* ) ) OR ( TITLE-ABS-KEY ( continu* ) ) OR ( TITLE-ABS-KEY (
follow-through* ) ) OR ( TITLE-ABS-KEY ( persist* ) ) OR ( TITLE-ABS-KEY ( engagement* ) )
OR ( TITLE-ABS-KEY ( refusal* ) ) OR ( TITLE-ABS-KEY ( non-treatment* ) ) OR (
TITLE-ABS-KEY ( non AND treatment* ) ) OR ( TITLE-ABS-KEY ( nontreatment* ) ) OR (
TITLE-ABS-KEY ( &apos;&apos;return AND to AND use&apos;&apos; ) ) )
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CINAHL	163
1. MH (“Artificial Intelligence+” OR “Decision Trees+” OR “Data Mining+”) 2. AB (“artificial intelligence*” OR “neural network*” OR “Support Vector Machine*” OR “machine learning” OR “Deep Learning” OR “Decision Tree*” OR “Data Mining” OR “text mining” OR “Naive bayes” OR “k-nearest neighbor*” OR “random forest*” OR “gradient boosting” OR “adaptive boosting” OR “regression tree*” OR “Natural Language Processing” OR “Automated reasoning” OR “Predictive analytic*” OR “Supervised learning” OR “Unsupervised learning”) 3. AB ((“Prediction model*” OR “predictive model*”) N4 (algorithm OR AI OR ML OR DL OR “logistic regression”)) 4. TI (“artificial intelligence*” OR “neural network*” OR “Support Vector Machine*” OR “machine learning” OR “Deep Learning” OR “Decision Tree*” OR “Data Mining” OR “text mining” OR “Naive bayes” OR “k-nearest neighbor*” OR “random forest*” OR “gradient boosting” OR “adaptive boosting” OR “regression tree*” OR “Natural Language Processing” OR “Automated reasoning” OR “Predictive analytic*” OR “Supervised learning” OR “Unsupervised learning”) 5. TI ((“Prediction model*” OR “predictive model*”) N4 (algorithm OR AI OR ML OR DL OR “logistic regression”))	

6. MH ("Opium+" OR "Narcotics+" OR "Substance Use Disorders+" OR "Opiate Overdose" OR "Opioid Epidemic" OR "Narcotic Antagonists+")

7. AB (Opioid* OR Opiat* OR Avinza* OR Buprenorphine* OR Codeine* OR Fentanyl* OR Heroin* OR Hydrocodone* OR Kadian* OR Methadone* OR Morphine* OR Naltrexone* OR Narcot* OR naloxone* OR Opana* OR Opium* OR Oxycodone* OR Oxymorphone* OR OxyContin* OR Percocet* OR Suboxone* OR Subutex OR Vivitrol* OR Vicodin* OR “Substance use*” OR “substance dependen*” OR “Substance disorder*” OR “Substance abuse*” OR “Substance misuse”)

8. TI (Opioid* OR Opiat* OR Avinza* OR Buprenorphine* OR Codeine* OR Fentanyl* OR Heroin* OR Hydrocodone* OR Kadian* OR Methadone* OR Morphine* OR Naltrexone* OR Narcot* OR naloxone* OR Opana* OR Opium* OR Oxycodone* OR Oxymorphone* OR OxyContin* OR Percocet* OR Suboxone* OR Subutex OR Vivitrol* OR Vicodin* OR “Substance use*” OR “substance dependen*” OR “Substance disorder*” OR “Substance abuse*” OR “Substance misuse”)

9. MH ("Medication Compliance" OR "Patient Compliance+" OR "Patient Dropouts")

10. AB (Adher* OR nonadher* OR non-adher* OR non adher* OR discontinu* OR complian* OR Noncomplian* OR non-complian* OR non complian* OR nonappear* OR non-appear* OR dropout* OR drop-out* OR non-attend* OR nonattend* OR attend* OR noshow* OR no-show* OR absent* OR absence* OR retention* OR retain* OR attrition* OR continu* OR follow-through* OR persist* OR engagement* OR refusal* OR non-treatment* or non treatment* OR nontreatment* OR “return to use”)

11. TI (Adher* OR nonadher* OR non-adher* OR non adher* OR discontinu* OR complian* OR Noncomplian* OR non-complian* OR non complian* OR nonappear* OR non-appear* OR dropout* OR drop-out* OR non-attend* OR nonattend* OR attend* OR noshow* OR no-show* OR absent* OR absence* OR retention* OR retain* OR attrition* OR continu* OR follow-through* OR persist* OR engagement* OR refusal* OR non-treatment* or non treatment* OR nontreatment* OR “return to use”)

12. 1 OR 2 OR 3 OR 4 OR 5

13. 6 OR 7 OR 8

14. 9 OR 10 OR 11

15. 12 AND 13 AND 14

Web of Science

795

1. (TS=(“artificial intelligence” OR “Neural Network*” OR “Support Vector Machine*” OR “machine learning” OR “Deep Learning” OR “Decision Tree*” OR “Data Mining” OR “text mining” OR “Naive bayes” OR “k-nearest neighbor*” OR “random forest*” OR “gradient boosting” OR “regression tree*” OR “adaptive boosting” OR “Natural Language Processing” OR “Automated reasoning” OR “Predictive analytic*”) OR TS=((“Prediction model” OR “predictive model”) NEAR/4 (algorithm OR AI OR ML OR DL OR “logistic regression”)) OR TS=(“supervised learning”) OR TS=(“unsupervised learning”))

2. (TS=(Opioid* OR Opiat* OR Avinza* OR Buprenorphine* OR Codeine* OR Fentanyl* OR Heroin* OR Hydrocodone* OR Kadian* OR Methadone* OR Morphine* OR Naltrexone* OR Narcot* OR naloxone* OR Opana* OR Opium* OR Oxycodone* OR Oxymorphone* OR OxyContin* OR Percocet* OR Suboxone* OR Subutex OR Vivitrol* OR Vicodin* OR Substance use* OR substance dependen* OR Substance disorder* OR Substance abuse* OR Substance misuse*))

3. (TS=(Adher* OR nonadher* OR non-adher* OR non adher* OR discontinu* OR complian* OR Noncomplian* OR non-complian* OR non complian* OR nonappear* OR non-appear* OR dropout* OR drop-out* OR non-attend* OR nonattend* OR attend* OR noshow* OR no-show* OR absent* OR absence* OR retention* OR retain* OR attrition* OR continu* OR follow-through* OR persist* OR engagement* OR refusal* OR non-treatment* or non treatment* OR nontreatment* OR “return to use”))

(TS=(“artificial intelligence” OR “Neural Network*” OR “Support Vector Machine*” OR “machine learning” OR “Deep Learning” OR “Decision Tree*” OR “Data Mining” OR “text mining” OR “Naive bayes” OR “k-nearest neighbo\$” OR “random forest*” OR “gradient boosting” OR “regression tree*” OR “adaptive boosting” OR “Natural Language Processing” OR “Automated reasoning” OR “Predictive analytic*”) OR TS=((“Prediction model” OR “predictive model”) NEAR/4 (algorithm OR AI OR ML OR DL OR “logistic regression”)) OR TS=(“supervised learning”) OR TS=(“unsupervised learning”)) AND (TS=(Opioid* OR Opiat* OR Avinza* OR Buprenorphine* OR Codeine* OR Fentanyl* OR Heroin* OR Hydrocodone* OR Kadian* OR Methadone* OR Morphine* OR Naltrexone* OR Narcot* OR naloxone* OR Opana* OR Opium* OR Oxycodone* OR Oxymorphone* OR OxyContin* OR Percocet* OR Suboxone* OR Subutex OR Vivitrol* OR Vicodin* OR Substance use* OR substance dependen* OR Substance disorder* OR Substance abuse* OR Substance misuse*)) AND (TS=(Adher* OR nonadher* OR non-adher* OR non adher* OR discontinu* OR complian* OR Noncomplian* OR non-complian* OR non complian* OR nonappear* OR non-appear* OR dropout* OR drop-out* OR non-attend* OR nonattend* OR attend* OR noshow* OR no-show* OR absent* OR absence* OR retention* OR retain* OR attrition* OR continu* OR follow-through* OR persist* OR engagement* OR refusal* OR non-treatment* or non treatment* OR nontreatment* OR “return to use”))

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"artificial intelligence" OR "neural network*" OR "support vector machine*" OR "machine learning" OR "deep learning" OR "decision tree*" OR "data mining" OR "text mining" OR "naive bayes" OR "k-nearest neighbo*" OR "random forest*" OR "gradient boosting" OR "regression tree*" OR "adaptive boosting*" OR "natural language processing" OR "automated reasoning" OR "predictive analytic*" OR "supervised learning" OR "unsupervised learning"

Opioid* OR Opiat* OR Avinza* OR Buprenorphine* OR Codeine* OR Fentanyl* OR Heroin* OR Hydrocodone* OR Kadian* OR Methadone* OR Morphine* OR Naltrexone* OR Narcot* OR naloxone* OR Opana* OR Opium* OR Oxycodone* OR Oxymorphone* OR OxyContin* OR Percocet* OR Suboxone* OR Subutex OR Vivitrol* OR Vicodin* OR "Substance use*" OR "substance dependen*" OR "Substance disorder*" OR "Substance abuse*" OR "Substance misuse*"

Adher* OR nonadher* OR non-adher* OR non adher* OR discontinu* OR complian* OR Noncomplian* OR non-complian* OR non complian* OR nonappear* OR non-appear* OR dropout* OR drop-out* OR non-attend* OR nonattend* OR attend* OR noshow* OR no-show* OR absent* OR absence* OR retention* OR retain* OR attrition* OR continu* OR follow-through* OR persist* OR engagement* OR refusal* OR non-treatment* or non treatment* OR nontreatment* OR "return to use"

190 Results for: *[[All: "artificial intelligence"] OR [All: "neural network*"] OR [All: "support vector machine*"] OR [All: "machine learning"] OR [All: "deep learning"] OR [All: "decision tree*"] OR [All: "data mining"] OR [All: "text mining"] OR [All: "naive bayes"] OR [All: "k-nearest neighbo*"] OR [All: "random forest*"] OR [All: "gradient boosting"] OR [All: "regression tree*"] OR [All: "adaptive boosting*"] OR [All: "natural language processing"] OR [All: "automated reasoning"] OR [All: "predictive analytic*"] OR [All: "supervised learning"] OR [All: "unsupervised learning"]]] AND [[All: opioid*] OR [All: opiat*] OR [All: avinza*] OR [All: buprenorphine*] OR [All: codeine*] OR [All: fentanyl*] OR [All: heroin*] OR [All: hydrocodone*] OR [All: kadian*] OR [All: methadone*] OR [All: morphine*] OR [All: naltrexone*] OR [All: narcot*] OR [All: naloxone*] OR [All: opana*] OR [All: opium*] OR [All: oxycodone*] OR [All: oxymorphone*] OR [All: oxycontin*] OR [All: percocet*] OR [All: suboxone*] OR [All: subutex] OR [All: vivitrol*] OR [All: vicodin*] OR [All: "substance use*"] OR [All: "substance dependen*"] OR [All: "substance disorder*"] OR [All: "substance abuse*"] OR [All: "substance misuse*"]]] AND [[Abstract: adher*] OR [Abstract: nonadher*] OR [Abstract: non-adher*] OR [Abstract: non adher*] OR [Abstract: discontinu*] OR [Abstract: complian*] OR [Abstract: noncomplian*] OR [Abstract: non-complian*] OR [Abstract: non complian*] OR [Abstract: nonappear*] OR [Abstract: non-appear*] OR [Abstract: dropout*] OR [Abstract: drop-out*] OR [Abstract: non-attend*] OR [Abstract: nonattend*] OR [Abstract: attend*] OR [Abstract: noshow*] OR [Abstract: no-show*] OR [Abstract: absent*] OR [Abstract: absence*] OR [Abstract: retention*] OR [Abstract: retain*] OR [Abstract: attrition*] OR [Abstract: continu*] OR [Abstract: follow-through*] OR [Abstract: persist*] OR [Abstract: engagement*] OR [Abstract: refusal*] OR [Abstract: non-treatment* or non treatment*] OR [Abstract: nontreatment*] OR [Abstract: "return to use"]]]*

IEEE	170
"artificial intelligence" OR "neural network" OR "neural networks" OR "Support Vector Machine" OR "Support Vector Machines" OR "machine learning" OR "Deep Learning" OR "Decision Tree" OR "Decision Trees" OR "Data Mining" OR "text mining" OR "Naive bayes" OR "k-nearest neighbor" OR "k-nearest neighbour" OR "random forest" OR "random forests" OR "gradient boosting" OR "adaptive boosting" OR "regression tree" OR "regression trees" OR "Natural Language Processing" OR "Automated reasoning" OR "Predictive analytic" OR "Predictive analytics" OR "Supervised learning" OR "Unsupervised learning"	
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non treatment OR non treatments OR nontreatment OR nontreatments OR "return to use"

Showing 1-25 of 170 results for ("All Metadata":artificial intelligence OR "All Metadata":neural network OR "All Metadata":neural networks OR "All Metadata":Support Vector Machine OR "All Metadata":Support Vector Machines OR "All Metadata":machine learning OR "All Metadata":Deep Learning OR "All Metadata":Decision Tree OR "All Metadata":Decision Trees OR "All Metadata":Data Mining OR "All Metadata":text mining OR "All Metadata":Naive bayes OR "All Metadata":k-nearest neighbor OR "All Metadata":k-nearest neighbour OR "All Metadata":random forest OR "All Metadata":random forests OR "All Metadata":gradient boosting OR "All Metadata":adaptive boosting OR "All Metadata":regression tree OR "All Metadata":regression trees OR "All Metadata":Natural Language Processing OR "All Metadata":Automated reasoning OR "All Metadata":Predictive analytic OR "All Metadata":Predictive analytics OR "All Metadata":Supervised learning OR "All Metadata":Unsupervised learning") AND ("All Metadata":Opioid OR "All Metadata":Opioids OR "All Metadata":Opiate OR "All Metadata":Opiates OR "All Metadata":Avinza OR "All Metadata":Avinzas OR "All Metadata":Buprenorphine OR "All Metadata":Buprenorphines OR "All Metadata":Codeine OR "All Metadata":Codeines OR "All Metadata":Fentanyl OR "All Metadata":Fentanyls OR "All Metadata":Heroin OR "All Metadata":Heroin OR "All Metadata":Hydrocodone OR "All Metadata":Hydrocodones OR "All Metadata":Kadian OR "All Metadata":Kadians OR "All Metadata":Methadone OR "All Metadata":Methadones OR "All Metadata":Morphine OR "All Metadata":Morphines OR "All Metadata":Naltrexone OR "All Metadata":Naltrexones OR "All Metadata":Narcotic OR "All Metadata":Narcotics OR "All Metadata":naloxone OR "All Metadata":naloxones OR "All Metadata":Opana OR "All Metadata":Opanas OR "All Metadata":Opium OR "All Metadata":Opiums OR "All Metadata":Oxycodone OR "All Metadata":Oxycodones OR "All Metadata":Oxymorphone OR "All Metadata":Oxymorphones OR "All Metadata":OxyContin OR "All Metadata":OxyContin OR "All Metadata":Percocet OR "All Metadata":Percocetamol OR "All Metadata":Suboxone OR "All Metadata":Suboxones OR "All Metadata":Subutex OR "All Metadata":Vivitrol OR "All Metadata":Vivitrols OR "All Metadata":Vicodin OR "All Metadata":Vicodins OR "All Metadata":Substance use OR "All Metadata":Substance user OR "All Metadata":Substance users OR "All Metadata":substance dependent OR "All Metadata":substance dependence OR "All Metadata":Substance disorder OR "All Metadata":Substance disorders OR "All Metadata":Substance abuse OR "All Metadata":Substance abuser OR "All Metadata":Substance abusers OR "All Metadata":Substance misuse OR "All Metadata":Substance misuser OR "All Metadata":Substance misusers") AND ("All Metadata":Adher* OR "All Metadata":nonadher* OR "All Metadata":non-adher* OR "All Metadata":non adhere* OR "All Metadata":discontinue OR "All Metadata":Discontinuation OR "All Metadata":compliant OR "All Metadata":Compliance OR "All Metadata":Noncompliant OR "All Metadata":Noncompliance OR "All Metadata":non-compliant OR "All Metadata":non-compliance OR "All Metadata":non compliant OR "All Metadata":non compliance OR "All Metadata":nonappear OR "All Metadata":non-appear OR "All Metadata":dropout OR "All Metadata":dropouts OR "All Metadata":drop-out OR "All Metadata":drop-outs OR "All Metadata":non-attend OR "All Metadata":non-attendance OR "All Metadata":nonattend OR "All Metadata":nonattendance OR "All Metadata":attend OR "All Metadata":attendance OR "All Metadata":noshow OR "All Metadata":nonshows OR "All Metadata":no-show OR "All Metadata":non

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Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) Checklist

SECTION	ITEM	PRISMA-ScR CHECKLIST ITEM	REPORTED ON PAGE #
TITLE			
Title	1	Identify the report as a scoping review.	1
ABSTRACT			
Structured summary	2	Provide a structured summary that includes (as applicable): background, objectives, eligibility criteria, sources of evidence, charting methods, results, and conclusions that relate to the review questions and objectives.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known. Explain why the review questions/objectives lend themselves to a scoping review approach.	4
Objectives	4	Provide an explicit statement of the questions and objectives being addressed with reference to their key elements (e.g., population or participants, concepts, and context) or other relevant key elements used to conceptualize the review questions and/or objectives.	4
METHODS			
Protocol and registration	5	Indicate whether a review protocol exists; state if and where it can be accessed (e.g., a Web address); and if available, provide registration information, including the registration number.	4 (https://osf.io/uzbhc/)
Eligibility criteria	6	Specify characteristics of the sources of evidence used as eligibility criteria (e.g., years considered, language, and publication status), and provide a rationale.	5
Information sources*	7	Describe all information sources in the search (e.g., databases with dates of coverage and contact with authors to identify additional sources), as well as the date the most recent search was executed.	4
Search	8	Present the full electronic search strategy for at least 1 database, including any limits used, such that it could be repeated.	Suppl 1
Selection of sources of evidence†	9	State the process for selecting sources of evidence (i.e., screening and eligibility) included in the scoping review.	4
Data charting process‡	10	Describe the methods of charting data from the included sources of evidence (e.g., calibrated forms or forms that have been tested by the team before their use, and whether data charting was done independently or in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought and any assumptions and simplifications made.	5
Critical appraisal of individual sources of evidence§	12	If done, provide a rationale for conducting a critical appraisal of included sources of evidence; describe the methods used and how this information was used in any data synthesis (if appropriate).	N/A
Synthesis of results	13	Describe the methods of handling and summarizing the data that were charted.	5

SECTION	ITEM	PRISMA-ScR CHECKLIST ITEM	REPORTED ON PAGE #
RESULTS			
Selection of sources of evidence	14	Give numbers of sources of evidence screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally using a flow diagram.	5
Characteristics of sources of evidence	15	For each source of evidence, present characteristics for which data were charted and provide the citations.	Table 1/2
Critical appraisal within sources of evidence	16	If done, present data on critical appraisal of included sources of evidence (see item 12).	N/A
Results of individual sources of evidence	17	For each included source of evidence, present the relevant data that were charted that relate to the review questions and objectives.	Table 1/2
Synthesis of results	18	Summarize and/or present the charting results as they relate to the review questions and objectives.	6-7
DISCUSSION			
Summary of evidence	19	Summarize the main results (including an overview of concepts, themes, and types of evidence available), link to the review questions and objectives, and consider the relevance to key groups.	7-8
Limitations	20	Discuss the limitations of the scoping review process.	9
Conclusions	21	Provide a general interpretation of the results with respect to the review questions and objectives, as well as potential implications and/or next steps.	9
FUNDING			
Funding	22	Describe sources of funding for the included sources of evidence, as well as sources of funding for the scoping review. Describe the role of the funders of the scoping review.	10

JB1 = Joanna Briggs Institute; PRISMA-ScR = Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews.

* Where *sources of evidence* (see second footnote) are compiled from, such as bibliographic databases, social media platforms, and Web sites.

† A more inclusive/heterogeneous term used to account for the different types of evidence or data sources (e.g., quantitative and/or qualitative research, expert opinion, and policy documents) that may be eligible in a scoping review as opposed to only studies. This is not to be confused with *information sources* (see first footnote).

‡ The frameworks by Arksey and O'Malley (6) and Levac and colleagues (7) and the JB1 guidance (4, 5) refer to the process of data extraction in a scoping review as data charting.

§ The process of systematically examining research evidence to assess its validity, results, and relevance before using it to inform a decision. This term is used for items 12 and 19 instead of "risk of bias" (which is more applicable to systematic reviews of interventions) to include and acknowledge the various sources of evidence that may be used in a scoping review (e.g., quantitative and/or qualitative research, expert opinion, and policy document).

From: Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. *Ann Intern Med*. ;169:467–473. doi: 10.7326/M18-0850